Experimental methods in psychological sciences

Course for ENS-PSL cognitive science masters program

Part 2: Psychophysics and signal detection theory

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Section 3. Scales and appearance

Last week we examined psychometric functions, which describe how an observer's choices (detection, identification, discrimination) change as a function of an externally manipulated stimulus property. The psychometric function plots the proportion of responses against the external stimulus property value, and describes the shape based on additive internal noise and a response criterion (just like Signal Detection Theory).

We also saw several ways observers responses can deviate from the predictions of the standard psychometric function:

1. **Task related factors**: lapses of attention and non-stable criteria can make it look like there is more noise adding variance to the psychological experience of the stimulus property than there really is.
2. **'Appearance' factors**: the magnitude or quality of the psychological experience doesn't vary linearly with the scale of the externally presented stimulus property.

With enough trials, we can see systematic deviations from the psychometric function and account for these factors in our measurements. But so far, we have been focussed on measuring sensitivity (how much noise). What if we don't just want to account for appearance factors, but we actually want to measure how things appear? What is a psychological experience 'like'? (Nagel, 1980).

One way to measure 'what it is like', is to measure the relative similarity to something else. And one way to measure the relative similarity is to measure how different something is.

a. The Just Noticeable Difference (JND)

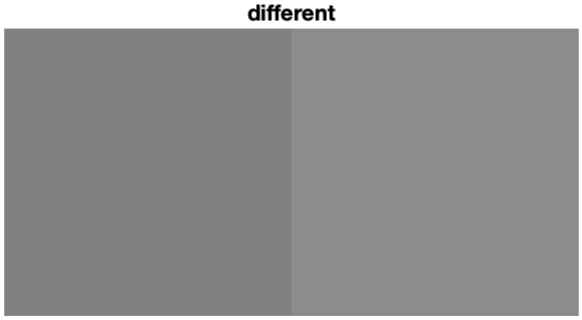
The Just Noticeable Difference (JND) measures the amount a stimulus property must be changed in order to notice a difference in that property on a certain proportion of trials. The proportion of trials is usually 0.5 or 0.75, depending on the task design and the choice of analysis. The stimulus property of interest has traditionally been some sort of magnitude, such as brightness or loudness (or monetary value). The measurement is equivalent to finding the 'threshold' of the psychometric function, as we did last week. You might hear people talking about comparing thresholds more broadly, the analysis and underlying theory is normally the same as what we will simulate here for JNDs.

Note that, unlike a recent interpretation (Sanford & Halberda, 2022), the JND was never meant to imply that below the threshold/JND-value, no difference is perceivable. The JND has always measured the value at which a difference is *reliably reported*, based on a certain proportion of trials.

I use the term *reliably reported* to highlight again, as we saw last week, the measurement of the threshold/JND can be affected by bias.

How is the JND measured? Here we use a task where the observer detects a difference between two stimuli. A simple example is brightness perception: one can present two squares of similar luminance and ask if they are the same or different. A visualisation:

```
% one square at 50% grey, another at 51% grey, probably not so perceivable
% difference

grey = ones(300,300)*0.5;
greyplus1 = ones(300,300)*0.51;

similar = [grey,greyplus1];

% another square at 55% grey should be clearly different (depending on your
% screen brightness

greyplus5 = ones(300,300)*0.55;
dissimilar = [grey,greyplus5];

figure;
subplot(2,1,1)
imshow(similar)
title('Similar')
subplot(2,1,2)
imshow(dissimilar)
title('different')
```



Although two stimuli are presented, this is not necessarily a forced-choice task, because the observer is making a single comparison between the two. A two-alternative forced-choice version would be if both the similar and dissimilar squares were presented on each trial and the observer had to choose which was more different.

TASK: Turn your screen brightness up until you see a difference in the similar squares. Turn your screen brightness down until you don't see a difference in the similar squares.

The JND for brightness depends on:

1. The difference in the stimulus intensities, termed *delta-I*
2. The base intensity to which this difference is added, often termed the 'pedestal', or just *I* (for intensity) (which is what you adjusted when you changed your screen brightness)

The JND is relative to the pedestal as well as the difference. This is not just the case for brightness, but for many other properties. This is attributed to the fact that the perceived magnitude is not linearly equivalent to the presented magnitude. In this way, JNDs can be used to measure something about appearance, the perceived magnitude, by essentially accumulating them (Luce & Edwards, 1958).

The general equation is that the perceived value, p, is some function of the physical stimulus value, f(x):

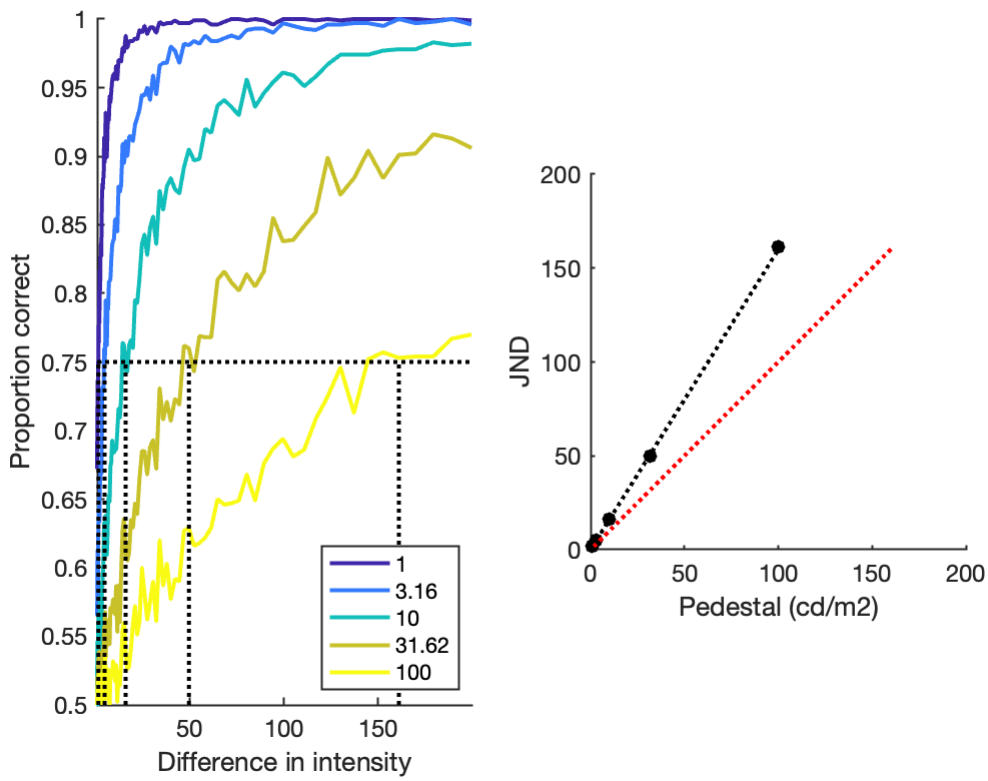
$p = f(x)$

The question is, what is the function f(x)? This question tells us something about 'what it is like'.

A demonstration:

```
% The pedestal luminance, in candelas per metre squared (cd/m2)
nps = 5;
pedestals = logspace(0,2,nps); % this gives log-spaced numbers

% The different difference values to test against
nds = 100;
delta_I = logspace(0,2.3,nds);

% Let's say theres a logarithmic relation between candelas and perceived
% brightness
perceived_brightness = log(pedestals); % logarithmic relationship
%perceived_brightness = (pedestals); % linear relationship
%perceived_brightness = (pedestals.^0.7); % power function

% this is affected by a constant value of internal noise
noise = 1;

% For each pedestal, simulate the psychometric function across difference
% values (2AFC design)
pfs = nan(nds,nps);

% many simulated trials
nsim = 1000;

% to save time, use the same simulated noise over and over again
i1_noise = randn(nds,nsim).*noise; % interval 1
i2_noise = randn(nds,nsim).*noise;


for npi = 1:nps

    % for efficiency, put the pedestal in the first interval
    % normally this would be randomly allocated
    i1_p = i1_noise + perceived_brightness(npi);

    % need to compute the perception of the pedestal+delta_I
    test_lum = delta_I + pedestals(npi);
    test_bright = log(test_lum);
    %test_bright = (test_lum);
    %test_bright = (test_lum.^0.7);

    i2_p = i2_noise + test_bright';

    % the psychometric function is the proportion choose i2 as greater
    % which is also proportion correct here (because the difference is
    % always added to i2, not like in a real experiment)
    pfs(:,npi) = mean(i2_p>i1_p,2);

end

% find the JND
% rather than fitting a psychometric funtion, let's just smooth out the
% curve and get the first point above 0.75
```

```
% this is just a shortcut, because we've sampled the psychometric function
% so finely

smooth_pf = movmean(pfs,5,1);
[~,jnd_inds] = max(smooth_pf>0.75,[],1);
jnds = delta_I(jnd_inds);

% plot
figure;
subplot(1,2,1)
colours = parula(nps);
legendEntry = cell(1,nps);
hold on
for npi = 1:nps

    plot(delta_I,pfs(:,npi),'Color',colours(npi,:),'LineWidth',2)
    legendEntry{npi} = num2str(round(pedestals(npi),2));

end

plot(delta_I,ones(size(delta_I))*0.75,'k:','LineWidth',2)

for ji = 1:nps

    plot([jnds(ji),jnds(ji)],[0.5,0.75],'k:','LineWidth',2)

end

axis([-inf,inf,0.5,1])
legend(legendEntry,'location','southeast')
xlabel('Difference in intensity')
ylabel('Proportion correct')
makeMyPlotPretty


subplot(1,2,2)
hold on
scatter(pedestals,jnds,50,'ko','filled')

weberConstant = mean(jnds./pedestals);
plot(pedestals,pedestals.*weberConstant,'k:','LineWidth',2)
plot(jnds,jnds,'r:','LineWidth',2)
axis square

xlabel('Pedestal (cd/m2)')
ylabel('JND')
makeMyPlotPretty
```



The figure on the left shows the difference in intensity by proportion correct, which is used to estimate the JND (in this case, 75% correct). The JND plotted against the pedestal on the right shows a linear relationship between JND and pedestal: as the pedestal increases, the JND increases. The red dashed line shows equality, the black dashed line shows a slope equivalent to "Weber's constant". Fechner, a student of Weber, described the principles behind this in 1860. Fechner termed "Weber's law" the equation (now frequently called the Weber-Fechner law):

delta_I/I = k

Where k is Weber's constant. This law suggests that for any stimulus intensity (I), the just noticeable difference (delta_I) is proportional to some constant, k.

We see this in the simulation above, when we simulate an observer whose percieved intensity is the logarithm of the actual intensity. The logarithm can be mathematically obtained from Weber's law with some awkward assumptions.

delta_p = k*delta_I/I

The difference in perception is equal to Weber's constant (k) times the difference in intensity relative to the intensity. The next awkward step is to say that anything true of a difference as small as a JND should also be true of smaller differences - the differential:

*d*p = k*_d_I/I

Integrating and solving you end up with

p = k*log(I/I0)

Where I0 is some threshold intensity. Luce & Edwards (1958) explain why this is wrong in principle, even though it often holds in practice. A rationale has also been put forward that the logarithmic relationship can be traced back to properties of neural firing rates.

More simply, for p = f(x), f(x) = log(x)

TAKE-AWAY: A linear relation between JNDs and pedestals is indicative (in some way) of a logarithmic relation between perceived intensity and actual intensity.

Had the observer's experience had a different functional relation to the stimulus intensity, f(x) = x (linear), or f(x) = x^b (power-law), the relation between the JND and the stimulus intensity would be different. For linear, the JND would be constant across stimulus intensity (stay the same value); for the power-law, the JND also follows a power function across stimulus intensity. I've put some code above that you can un-comment to check and visualise if you want (for the power law, increase the noise to 10 and don't plot the Weber constant prediction).

b. Measuring JNDs

Last week we saw that getting a good measure of a psychometric function requires lots of trials. Getting a good measure of the relation between stimulus intensity and the JND similarly requires lots of JNDs. So, if we measure JNDs with psychometric functions, we end up with a lot a lot of trials. A short-cut is available: use a staircase procedure.

The adaptive staircase adjusts the stimulus level trial-by-trial to achieve a certain proportion correct. There are sophisticated methods to do this (QUEST, which uses a Bayesian adaptive search, Watson & Pelli, 1983; ASA, Kesten, 1958), but the basic method of adjusting up and down based on the series of correct/incorrect responses (Cornsweet, 1962) is still most comonly used today. The idea is to 'converge' on the threshold value.

There are some tricks. As we will explore a bit later, perceptual interactions are common, meaning two stimuli in close spatial or temporal proximity will have an effect on the perception of each stimulus. With a staircase, the current stimulus value is strongly predicted by the previous, and that previous stimulus can affect the perception of the current one. I like to mix things up by interleaving staircases, one starting well above threshold, and one below, or better, two of each. You can also moderate the step-size over trials, to get a finer measure of the threshold quicker, but this can be dangerous if the staircase is in the wrong place when the step-size is decreased.

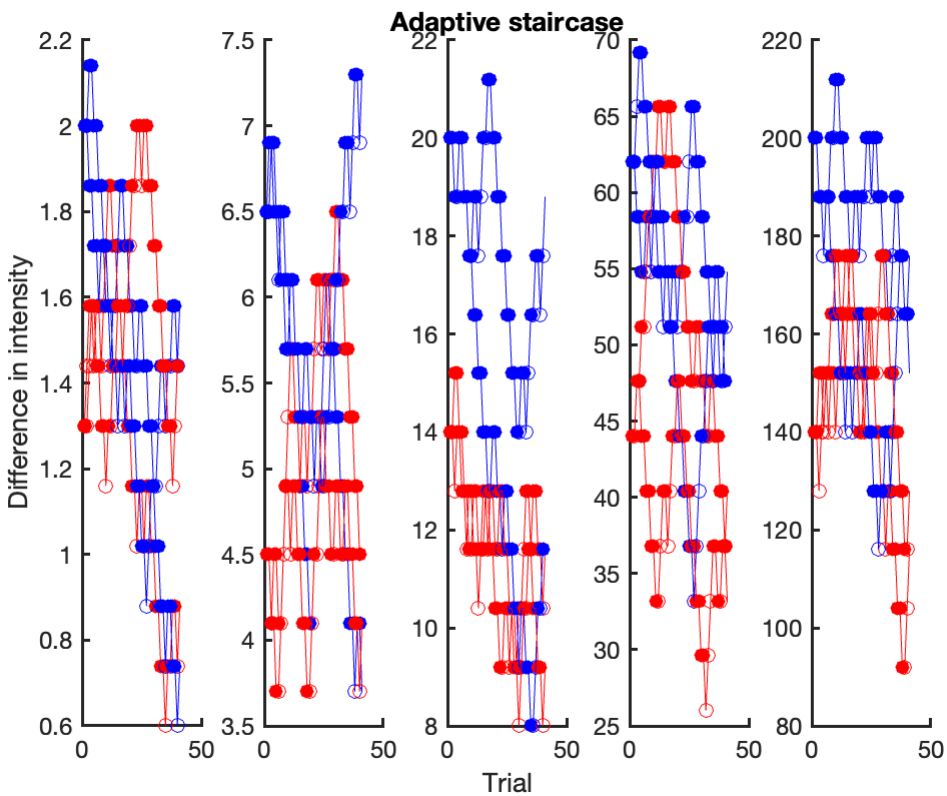
Let's go fairly basic, with two 2-down 1-up staircases, with a fixed number of trials.

```
% The pedestal luminance, in candelas per metre squared (cd/m2)
nps = 5;
pedestals = logspace(0,2,nps); % this gives log-spaced numbers

% Let's say theres a logarithmic relation between candelas and perceived
% brightness
perceived_brightness = log(pedestals); % logarithmic relationship
%perceived_brightness = (pedestals); % linear relationship
```

```
%perceived_brightness = (pedestals.^0.7); % power function

% this is affected by a constant value of internal noise
noise = 1;

% set the starting point of the staircase for each pedestal
% I'll choose it manually from the simulation above
% normally I would pilot the experiment to get approximate values
stair_start = [2,1.3;...
    6.5,4.5;...
    20,14;...
    62,44;...
    200,140];
step_size = (stair_start(:,1)-stair_start(:,2))./5;

ntrials = 40;

% perform a small number of simulations
nsim = 2;

jnds = nan(nps,nsim);

% for each pedestal, simulate the staircase
% plot the stairs as we go
figure;

for npi = 1:nps

    subplot(1,nps,npi)
    hold on
    makeMyPlotPretty

    % for each simulation
    for si = 1:nsim

        ss1 = nan(1,ntrials+1);
        ss2 = nan(1,ntrials+1);

        ss1(1) = stair_start(npi,1);
        ss2(1) = stair_start(npi,2);
        prevC1 = 0;
        prevC2 = 0;

        for ti = 1:ntrials

            % two stimuli
            i1_p = (randn(1,1)*noise) + perceived_brightness(npi);
            i2_p = (randn(1,1)*noise) + log(ss1(ti)+pedestals(npi));

            if i1_p>i2_p % an error, increase the stimulus value

                scatter(ti,ss1(ti),50,'bo');
                ss1(ti+1) = ss1(ti)+step_size(npi);
                prevC1 = 0;
```

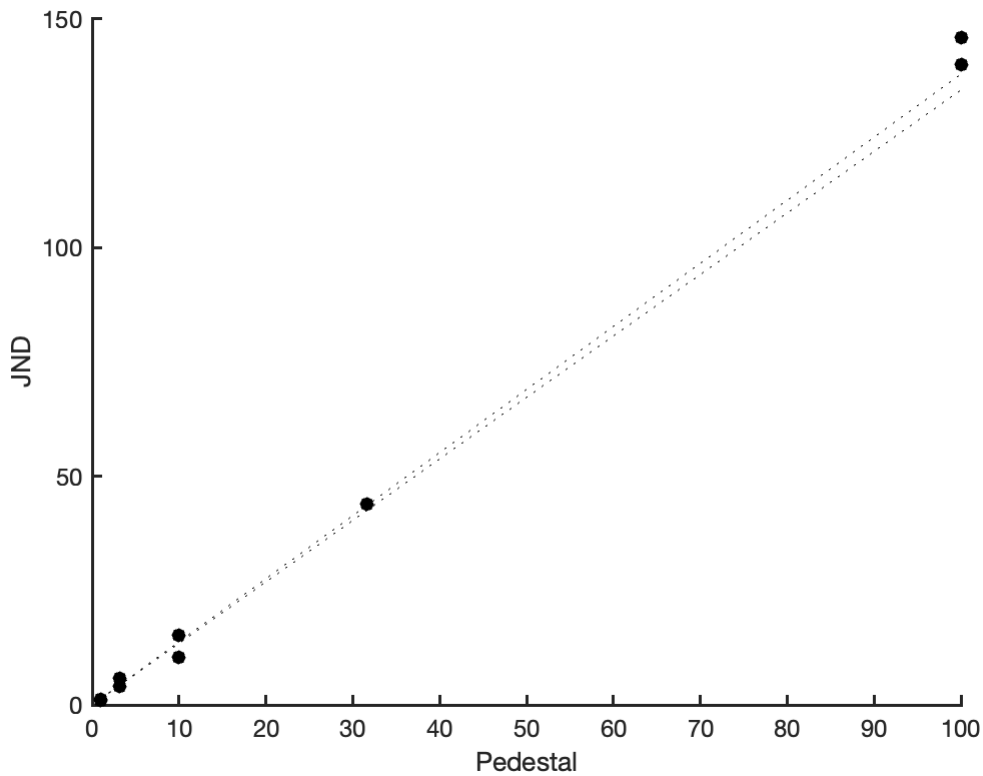
```
        else % correct

            scatter(ti,ss1(ti),50,'bo','filled')

            if prevC1 == 1 % decrease if that's two in a row

                if ss1(ti) == ss1(ti-1)
                    ss1(ti+1) = ss1(ti) - step_size(npi);
                    prevC1 = 1;
                else
                    ss1(ti+1) = ss1(ti);
                    prevC1 = 1;
                end

            else

                ss1(ti+1) = ss1(ti);
                prevC1 = 1;

            end

        end

        % and the same for the other interleaved staircase
        % two stimuli
        i1_p = (randn(1,1)*noise) + perceived_brightness(npi);
        i2_p = (randn(1,1)*noise) + log(ss2(ti)+pedestals(npi));

        if i1_p>i2_p % an error, increase the stimulus value

            scatter(ti,ss2(ti),50,'ro');
            ss2(ti+1) = ss2(ti)+step_size(npi);
            prevC2 = 0;

        else % correct

            scatter(ti,ss2(ti),50,'ro','filled')

            if prevC2 == 1 % decrease if that's two in a row

                if ss2(ti) == ss2(ti-1)
                    ss2(ti+1) = ss2(ti) - step_size(npi);
                    prevC2 = 1;
                else
                    ss2(ti+1) = ss2(ti);
                    prevC2 = 1;
                end

            else

                ss2(ti+1) = ss2(ti);
                prevC2 = 1;
```

```
                end

            end

        end

        plot(1:ntrials+1,ss1,'b')
        plot(1:ntrials+1,ss2,'r')

        jnds(npi,si) = ( ss1(end)+ss2(end) )/2;

    end
end
subplot(1,nps,round(nps/2))
xlabel('Trial')
title('Adaptive staircase')
subplot(1,nps,1)
ylabel('Difference in intensity')
```




```
figure;
hold on
scatter(pedestals,jnds,50,'ko','filled')
% calculate Weber constants
wc = nan(1,nsim);
for si = 1:nsim
    wc(si) = mean(jnds(:,si)./pedestals');
    plot(pedestals,pedestals*wc(si),'k:')
end
```

```
xlabel('Pedestal')
ylabel('JND')
makeMyPlotPretty
```



You can see from the above that even with 40 trials in each of 2 staircases, the measure is still a bit rough. That's where the more sophisticated staircase procedures help. A separate problem is that this simulated observer is pretty perfect for a staircase, but humans will have other issues, such as lapses, and biases, which will make this measurement even worse.

c. Why not just ask people what it's like?

To measure something so subjective, maybe a subjective measure is best. But how do we ask people to report what something is like in a measureable way? One method is the method of adjustment. In this method you show an observer a stimulus and then ask them to adjust another stimulus until it looks/sounds like the one they just saw. Usually you might play some trick with the first stimulus (put it in a different context, or mask it) to make this more interesting than just reproduction. Although the method of adjustment has been used very well in many studies, there are two problems with this method that are difficult to combat.

The first is the problem mentioned above (and dealt with more below) of perceptual interactions. As you adjust the stimulus you current percept is impacted by how it looked before. It's common that if a stimulus is adjusted

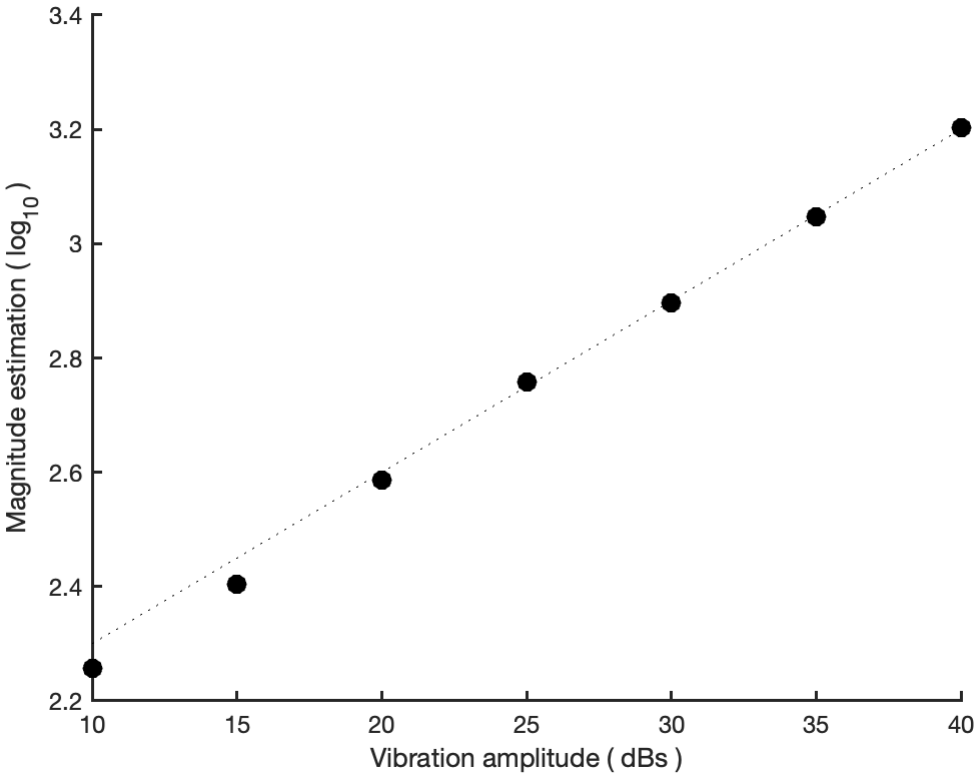
in one direction the report will be biased further in that direction (the observer turned the knob too far). This kind of bias can be accounted for by having both upward and downward adjustments.

The second problem is more theoretical. The El Greco fallacy is named after the fallacious suggestion that the elongated painting style of El Greco was due to a particular astigmatism, which made El Greco perceive the world more elongated. If El Greco had seen the world more elongated, and was painting the world as he saw it, then he would have needed to paint the world normally to perceive his own paintings as he perceives the world. With the method of adjustment, a perfect replication doesn't necessarily mean a perfect perception. An experiement needs to be carefully (and often, creatively) designed for the method to tell us something interesting about our inner perceptual experiences.

Another way to ask people what it's like is to ask them to simply report a number corresponding to their perceived magnitude. This became known as magnitude estimation, and was championed by Stevens (1957). In his post-humously published book, *Psychophysics* (1975; completed by his wife, Geraldine Stevens, who also participated in his experiements for over 30 years), Stevens describes his amazement that he could simply rescale observers' number responses and plot them in a nice straight line in log-log coordinates. This amazement stemmed from psychologists battling to have their measurement of psychological experiences recognised as genuine scientific measurement. The opinion at the time was that our perception results from a "*…slow and doubtless complicated process … liable to more or less accidental derangement at every step of its progress*." (Pierce & Jastrow, 1885): it was a waste of time to try to measure it at all.

We can simulate one of Stevens's experiments. For something different (not vision nor audition): observers held a thin metal metal rod between their thumb and forefinger as a vibration of a particular frequency and amplitude was applied. Stevens asked observers to report a number corresponding to the amplitude (perceived magnitude, or intensity) of the vibration.

```
% Stevens plots his data in dB, a logarithmic scale
dBs = 10:5:40;

% He reports sensation is a power function of amplitude
amplitudes = 10.^(dBs/20);
perceived_amplitude = amplitudes.^0.6;

% Add some Gaussian noise, there's always gaussian noise
perceived_amplitude = perceived_amplitude + randn(28,length(dBs));

% Observers report a number corresponding to their experience - any number,
% on any scale
estimate = round(perceived_amplitude*100);

% plot
figure;
hold on
scatter(dBs,log10(mean(estimate,1)),100,'k','filled')
plot(dBs, (dBs*0.03)+2,'k:')
xlabel('Vibration amplitude ( dBs )')
ylabel('Magnitude estimation ( log_1_0 )')
makeMyPlotPretty
```



Stevens's measurements genuinely look this neat. He had 14 observers make 2 estimates of each amplitude, totalling 28 estimates over 7 stimulus intensities, 196 data-points, as simulated. The messy plot of JNDs above simulated 400 trials over just 5 stimulus intensities. One can be impressed with how precise people are at reporting magnitude estimates.

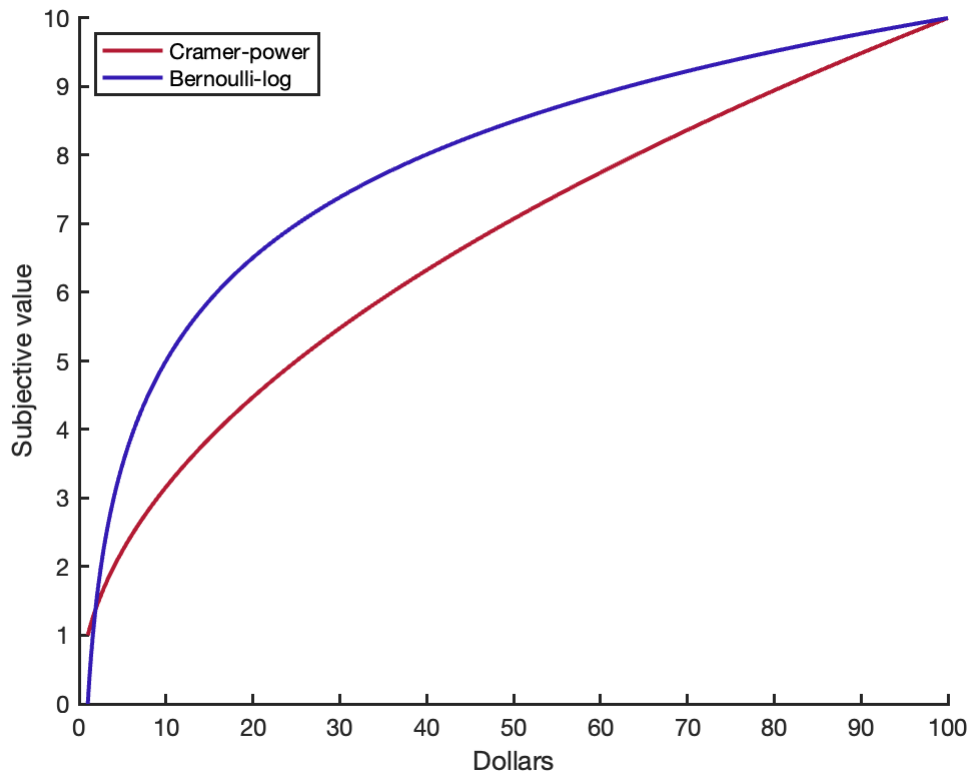
Based on this, and many more data-sets over an impressive variety of stimuli and senses, Stevens concluded there was a power-law relation between perceived intensity and stimulus intensity. This became known as Stevens' power law.

d. The big controversy

The Weber-Fechner law was meant to be a universal law for all sensation. It specified a logarithmic scale of perceived intensity. But Stevens consistently found a power function. This is not the first log vs power debate. In 1728 Cramer proposed a power function to describe the subjective value of money. Bernoulli proposed a logarithmic function in 1738. The two functions actually look quite similar:

```
Dollars = linspace(1,100,999);

% Cramer's power function
cramer_subjective_value = Dollars.^0.5;
```

```
% Bernoulli's log function
bernoulli_subjective_value = log10(Dollars)*5;

figure;
hold on
plot(Dollars,cramer_subjective_value,'Color',[0.7,0.1,0.2],'LineWidth',2)
plot(Dollars,bernoulli_subjective_value,'Color',[0.2,0.1,0.7],'LineWidth',2)
legend('Cramer-power','Bernoulli-log','location','northwest')
xlabel('Dollars')
ylabel('Subjective value')
makeMyPlotPretty
```



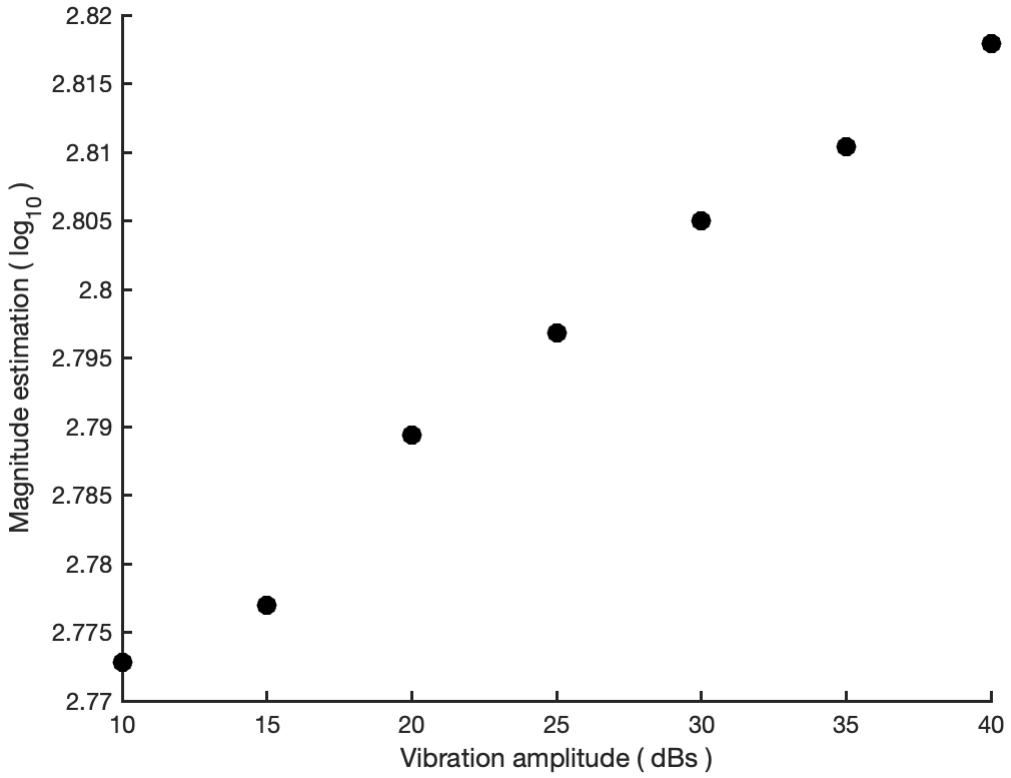
Both equations propose a steeper increase with small values that levels out as the value increases.

This wasn't considered particularly controversial in 1738. But in 1958, Luce and Edwards wrote "*The contraversy is particularly hot at present*...". This not only because Stevens was questioning the well established principles of Fechner (which could be considered the birth of psychophysics), but also within a context where the field of psychology was battling to establish itself as a quantitative science. In 1932 a committee (the Fergusson committee) had been appointed by the British Association for the Advancement of Science, composed of sections A, Mathematical and Physical Sciences, and J, Psychology, to "*consider and report upon the possibility of quantitative estimates of sensory events*". The final report was released in 1939, which detailed the inability of the committee to reconcile their views, one being that sensation intensities are not measurable. The response from the field of psychology was a rush of papers in the 40s and 50s attempting to legitimise psychology as a science. Questioning the Weber-Fechner law, showing that data aquired using a different measure of the same phenomenon leads to a different functional description, was not helping this

effort. Multiple papers were published to explain or rectify the differences between the Weber-Fechner law and Stevens' power law.

My favourite explanation is from Mackay, 1963. It's beautiful: Stevens missed a step. We missed a step. What we simulated above, we input a stimulus, it is transformed to a perceptual experience, then we just output the number. If we expect a transformation from stimulus-to-perception, why not expect another transformation from perception-to-response. The subjective value plot might also suggest that our perception of the maginitude of a number (the method of reporting) could also be non-linear. In fact, Mackay showed that applying two log functions (Weber's law on input and output) gets us back to Stevens power law.

```
% Back to Stevens's example experiment
dBs = 10:5:40;

% Now we simulate perception as a log of amplitude
amplitudes = 10.^(dBs/20);
perceived_amplitude = log(amplitudes);

% Add some Gaussian noise, there's always gaussian noise
perceived_amplitude = perceived_amplitude + randn(100,length(dBs));
perceived_amplitude = perceived_amplitude - min(min(perceived_amplitude)) + 1;
% to get rid of a problem below

% And a number has a log magnitude
response = log(perceived_amplitude*100);

% It's more realistic to give round number responses
estimate = (response*100); % just to give nice big numbers again

% plot
figure;
hold on
scatter(dBs,log10(mean(estimate,1)),100,'k','filled')
%plot(dBs, (dBs*0.03)+2,'k:')
xlabel('Vibration amplitude ( dBs )')
ylabel('Magnitude estimation ( log_1_0 )')
makeMyPlotPretty
```



Applying the log-law twice gets us back to the straight line that is indicative of a power-law. The power-law is not a description of stimulus-perception, but stimulus-perception-response.

KEY-TAKE-AWAY: For any subjective measure we need to consider *how* the observer makes their response.

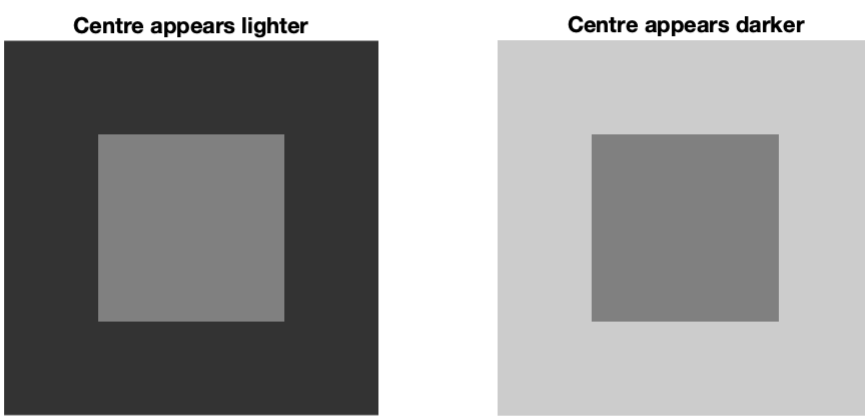
"It is an unavoidable aspect of psychological experiments that no matter how well instructed, it is the observer who determines how the task is done." Galvin et al., 2003

Optional extras

e. Beyond isolated magnitude: Maximum likelihood difference scaling

The experience of a sensation is not only a function of a stimulus property, but the context of that stimulus property. That is, our experience is not absolute, but relative. One immediate example of this is lightness constancy. The light that reaches our eyes is a function of both the reflectance of an object (a black object is reflecting less light than a white object) and the illumination (how much light is available to be reflected). When we turn off a light, we do not perceive all the objects in a room as changing to a darker grey, rather we perceive that there is less light in the room. This is lightness constancy, we interpret the change in the light reaching our eyes as a change in the overall luminance, not the reflectance of objects. One thing that helps us do this is the relative nature of perception. This can be easily demonstrated.

```
% one square at 50% grey, surrounded by a dark grey background

grey = ones(300,300)*0.5;
darkgrey = ones(600,600)*0.2;

stim1 = darkgrey;
stim1(151:450,151:450) = grey;

% another square exactly the same, surrounded by a light grey background

lightgrey = ones(600,600)*0.8;

square2 = lightgrey;
square2(151:450,151:450) = grey;

figure;
subplot(1,2,1)
imshow(stim1)
title('Centre appears lighter')
subplot(1,2,2)
imshow(square2)
title('Centre appears darker')
```



The centre squares are exactly the same grey, but they appear different because of the surround.

A typical example of lightness constancy is the checkerboard illusion, where a shadow cast on the checkerboard results in a white square having the same light reflected as a black square, though the observer still perceives them as white and black.

How do we understand the effect of multiple stimulus properties on perceptual experience?

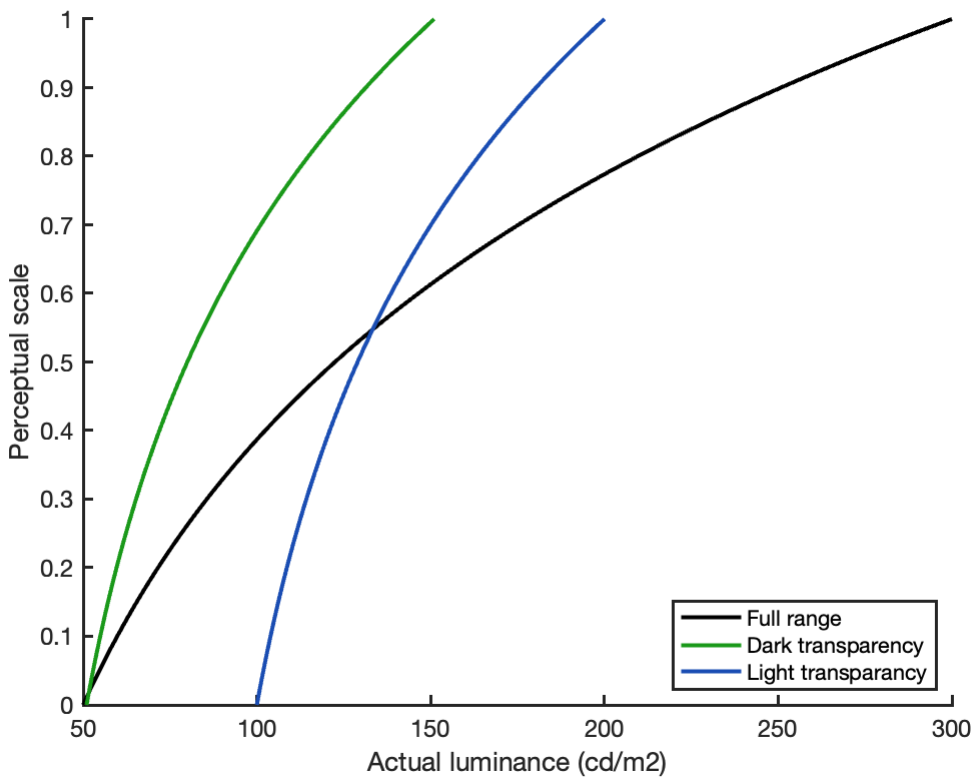
One recently developed method is maximum likelihood difference scaling (Maloney & Yang, 2003). This method asks observers to report the perceived difference between stimuli within a given context. A general linear model is then applied to retrieve the perceptual scale (Knoblauch & Maloney, 2008), and these scales can be compared across contexts.

Each trial the observer is presented with three (or four) stimuli and asked which pair is more different, the first two or the second two (for three stimuli, the centre is counted twice - is (x1 - x2) more different from (x2 - x3)). Aguilar & Maertens (2020) did this by presenting observers with checkerboard stimuli that appeared to be viewed through a dark or light transparent window.

To simulate lightness constancy, we assume the observer remaps the darkest thing to black and the lightest thing to white.

```
% normal viewing of the world, we could experience 50 to 300 candelas

worldLum = linspace(50,300,300);

% our perception is a log function of the actual luminance
perceivedLum = log(worldLum);

% normalise to 0 - 1
perceivedLum = perceivedLum - min(perceivedLum);
perceivedLum = perceivedLum ./ max(perceivedLum);


% the true stimulus without the added transparency varies in pixel
% luminance from 100 to 200 candelas

trueStimNoWindow = linspace(100,200,300);


% the stimulus with added dark transparency has luminance

trueStimDarkWindow = trueStimNoWindow-49;


% lightness constancy suggests the observer perceives the same
% lightness, despite the change in luminance
perceivedStimDarkWindow = perceivedLum;
perceivedStimLightWindow = perceivedLum;


figure;
hold on
plot(worldLum, perceivedLum,'k','LineWidth',2);
```

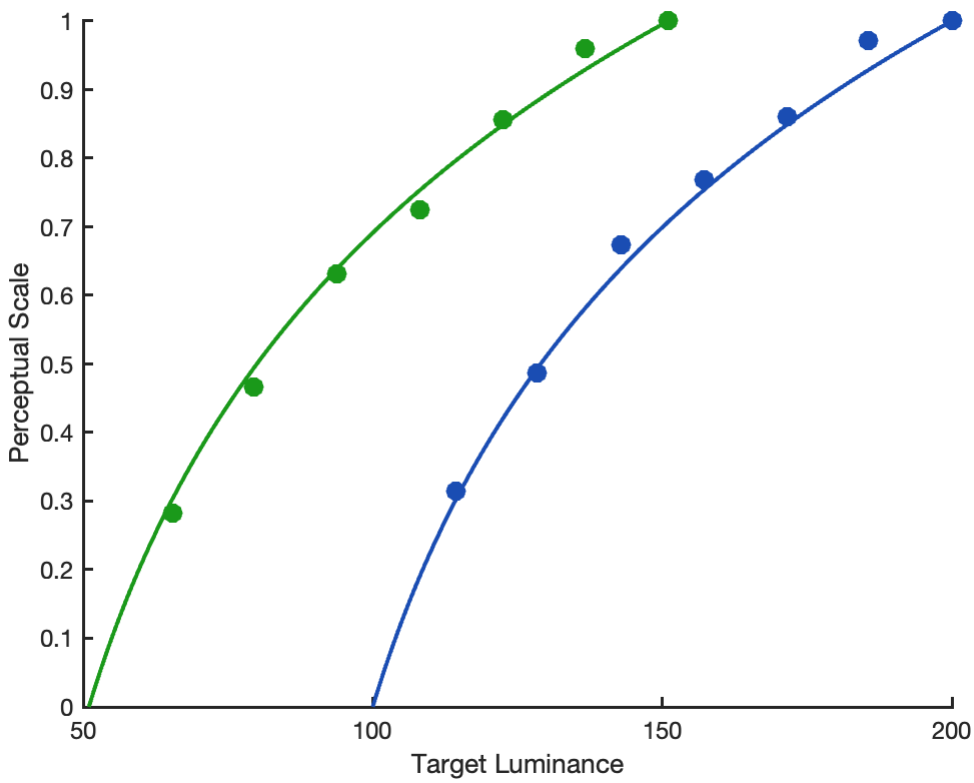
```
plot(trueStimDarkWindow,perceivedStimDarkWindow,'Color',[0.1,0.6,0.1],'LineWidth',2)
plot(trueStimLightWindow,perceivedStimLightWindow,'Color',[0.1,0.3,0.7],'LineWidth',2)

xlabel('Actual luminance (cd/m2)')
ylabel('Perceptual scale')

legend('Full range','Dark transparency','Light transparancy','location','southeast')
makeMyPlotPretty
```




```
% simulate responses of an observer in an MLDS experiment

% we take 8 stimuli to compare - normally we might choose these carefully
% to maximise power

stimInds = round(linspace(1,300,8));

theseStimDark = trueStimDarkWindow(stimInds);
theseStimLight = trueStimLightWindow(stimInds);

% the perceptual mean
thesePerceivedDark = perceivedStimDarkWindow(stimInds);
thesePerceivedLight = perceivedStimLightWindow(stimInds);

% perception is disrupted by noise
noiseSD = 0.1;

% take 10 trials of each combination
nreps = 10;
```

```
% the combination is 3 stimuli, in order
ntrials = ( 56 ) *nreps;

% pre-allocate
respsD = nan(ntrials,1);
respsL = nan(ntrials,1);
stims = zeros(ntrials,8);

ti = 1;

for si1 = 1:6

    for si2 = si1+1:7

        for si3 = si2+1:8

            for ri = 1:nreps

                p1 = thesePerceivedDark(si1)+(randn(1)*noiseSD);
                p2 = thesePerceivedDark(si2)+(randn(1)*noiseSD);
                p3 = thesePerceivedDark(si3)+(randn(1)*noiseSD);

                respsD(ti) = abs(p1 - p2) < abs(p2 - p3);

                p1 = thesePerceivedLight(si1)+(randn(1)*noiseSD);
                p2 = thesePerceivedLight(si2)+(randn(1)*noiseSD);
                p3 = thesePerceivedLight(si3)+(randn(1)*noiseSD);

                respsL(ti) = abs(p1 - p2) < abs(p2 - p3);

                stims(ti,si1) = 1;
                stims(ti,si2) = -2;
                stims(ti,si3) = 1;
                % why? the response is effectively
                % (p1 - p2) - (p2 - p1) < 0
                % = p1 - 2p2 + p3 < 0

                ti = ti + 1;


            end
        end
    end
end


% use a generalised linear model to get the perceptual scale
mdlSpec = 'y ~ x1 + x2 + x3 + x4 + x5 + x6 + x7 - 1';
mdlD = fitglm(stims(:,2:8),respsD,mdlSpec,'Distribution','binomial');

perceptualScaleD = mdlD.Coefficients.Estimate;
% normalise to 1
perceptualScaleD = perceptualScaleD./max(perceptualScaleD);
```

```
mdlL = fitglm(stims(:,2:8),respsL,mdlSpec,'Distribution','binomial');

perceptualScaleL = mdlL.Coefficients.Estimate;
perceptualScaleL = perceptualScaleL./max(perceptualScaleL);

figure;
hold on
scatter(theseStimDark(2:8),perceptualScaleD,100,[0.1,0.6,0.1],'filled')
scatter(theseStimLight(2:8),perceptualScaleL,100,[0.1,0.3,0.7],'filled')
plot(trueStimDarkWindow,perceivedStimDarkWindow,'Color',[0.1,0.6,0.1],'LineWidth',2)
plot(trueStimLightWindow,perceivedStimLightWindow,'Color',[0.1,0.3,0.7],'LineWidth',2)

xlabel('Target Luminance')
ylabel('Perceptual Scale')
makeMyPlotPretty
```



Using this method we recover the perceptual scale we modeled. This suggests this is a good method of measuring perceptual scales in different contexts. Maximum likelihood difference scaling has been used to measure the perceptual scale for many interesting properties: perceived lightness (Aguilar & Maertens, 2020), glossiness (Hansmann-Roth and Mamassian, 2017), color (Rogers et al., 2016), or the watercolor effect (Gerardin et al., 2014).

TAKE-AWAY: Psychophysics is full of methods to measure 'what it is like'. Some of these methods are still being developed. While they are mainly employed in research examining sensation, they can also be applied to all sorts of experiences, and in social psychology - to measure the perceived magnitude of a crime, for example.

Functions

```
function makeMyPlotPretty()

ax = gca;
ax.FontSize = 12;
ax.LineWidth=1.5;

end
```

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